

The Economic Effects of Flood Buyout Programs: Evidence from an Equilibrium Sorting Model

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Introduction

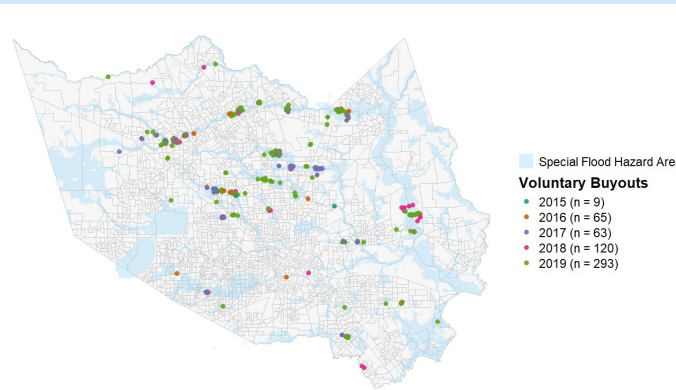
What is a Buyout Program?

A buyout program is government program that purchases flood-prone properties and converts the land to permanent open or green space, reducing future flood risk.

Research Questions:

1. Do households value living in neighborhoods with greater cumulative exposure to flood buyouts differently?
2. Do these responses differ across household groups, particularly by income and race/ethnicity?

Application



Data

Sample characteristics	Value
Low-income households	37.6%
Medium-income households	33.4%
High-income households	28.5%
White households	68.9%
Black households	10.6%
Asian households	8.3%
Hispanic households	33.1%
Annualized housing cost (\$) (mean/max)	11,925/96,688
Sale price (\$) (mean/max)	238,510/1,933,800
Voluntary buyout intensity (mean/max)	0.002/0.974
SFHA share (mean/max)	(0.094/1)

- Cotality: 64,905 housing transactions from Harris County, TX, 2015-19
- HMDA: Household income tercile and race/ethnicity
- US Census, GIS, & FEMA: Neighborhood characteristics and flood exposure
- Up to 2,830 neighborhoods in each year's choice set (varies annually)
- Harris Flood Control District: buyout locations and dates

Methods

Step 1: Recover Neighborhood Utility and Household Preferences

Households choose the neighborhood that maximizes utility:

$$V_{ijt} = \delta_{jt} + \sum_k \beta_k (Z_i \times B_{jt}) + \varepsilon_{ijt}$$

where:

- δ_{jt} : mean utility of neighborhood j in year t
- B_{jt} : cumulative voluntary buyout intensity
- Z_i : household characteristics (income, race/ethnicity)
- ε_{ijt} : household-specific preference shock

A nested maximum likelihood estimation (MLE) routine with a Berry (1994) contraction mapping recovers neighborhood mean utilities (δ_{jt}) by equating model-predicted and observed neighborhood choice shares. The estimation simultaneously recovers household-specific preference parameters (β_k) that measure how preferences for buyout exposure vary across demographic groups.

Step 2: Estimate Preferences for Buyout Exposure

Recovered neighborhood utilities from stage 1 are regressed on housing costs (P_{jt}), buyout intensity (B_{jt}), and neighborhood characteristics (X_{jt}):

$$\widehat{\delta}_{jt} = X_{jt}\beta + \alpha_P P_{jt} + \alpha_B B_{jt} + \varepsilon_{jt}$$

The coefficient (α_B) measures the utility (or preference) associated with living in neighborhoods with greater cumulative buyout intensity for the reference group (high-income White households). Stage 1 interaction estimates (β_k) capture how this preference differs for other demographic groups.

Identification: Housing costs are instrumented using the share of undeveloped land located 5–10 km away (Berry & Timmins, 2007) to address price endogeneity.

Step 3: Calculate Marginal Willingness to Pay

Marginal willingness to pay (MWTP) is calculated as

$$MWTP = \frac{\alpha_B + \beta_k}{-\alpha_P}$$

for each household group.

Future Directions & Policy Implications

- This work motivates studying neighborhood-scale mandatory buyout programs and their effects on residential location decisions.
- Future research will examine whether more coordinated buyouts generate different neighborhood responses than voluntary buyouts.
- Accounting for heterogeneous household preferences may improve the design of equitable flood adaptation policies.

Results

Finding 1: Buyout exposure is associated with lower neighborhood utility

Year	Coefficient	Std error
2015	-4.32***	0.64
2016	-1.60***	0.42
2017	-3.03***	0.56
2018	-5.52***	0.44
2019	-1.86***	0.59

Takeaway: Across every year from 2015–2019, neighborhoods with greater cumulative voluntary buyout exposure were consistently valued less.

Possible explanation: Buyouts may signal greater flood risk, neighborhood disruption, or uncertainty following repeated flooding.

Finding 2: Buyout preferences differ across household groups

Group	2018	2019
Low income	+4.91***	+0.68
Medium income	+5.08***	-1.34
Black	+2.30**	+1.79**
Hispanic	+1.66	+1.95***

Takeaway: Lower-income, Black, and Hispanic households showed weaker avoidance of buyout-intensive neighborhoods than high-income White households, especially after Hurricane Harvey.

Possible explanation: Housing affordability, fewer neighborhood options, different experiences with flood risk, different tradeoffs between amenities and housing costs.

How much are households willing to pay for buyout exposure?

Increase in cumulative buyout intensity	High-income White	Low-income	Black	Hispanic
1 percentage point	-\$420	-\$102	-\$354	-\$249
5 percentage points	-\$2,101	-\$510	-\$1,770	-\$1,245
10 percentage points	-\$4,202	-\$1,020	-\$3,540	-\$2,490
100 percentage points	-\$42,024	-\$10,204	-\$35,396	-\$24,901

Interpretation: A neighborhood where 10% of the 2010 housing stock has been bought out is valued about \$4,200 less by high-income White households than an otherwise similar neighborhood with no buyouts.

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